Nano Materials Engineer

TSMC · Austin, TX

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Employer	TSMC
Location	Austin, TX
Career stage	New grad / early career
Location group	Texas (outside DFW)
Major (archive tag)	Materials Science & Engineering
Sector area	Electronics & semiconductor manufacturing
Industry (LinkedIn)	Semiconductor Manufacturing
Pay listed	Not listed
Stated rules	No GPA or sponsorship rule stated
Posted / closes	2026-10-29 · 2027-01-19
Status when checked	Accepting applications on LinkedIn (Nov 1, 2026)
Source page	https://www.linkedin.com/jobs/view/4429624025/
Captured	2026-11-01

Full posting text as it appeared on LinkedIn on Nov 1, 2026, saved for ECS 1100. Postings change and close often; open the source page above for the current version before applying.

At a glance

What you would do

Engineer managing raw materials for semiconductor chip fabrication, solving material quality issues, running defect analyses and auditing suppliers.

Who can apply

- **Majors named:** Chemistry, Materials Science, Electrical Engineering, Chemical Engineering, Mechanical Engineering, Physics, Optics
- **Degree and experience:** Bachelor's, master's or PhD in Chemistry, Materials Science, EE, ChE, ME, Physics, Optics or related; semiconductor process experience a plus

Skills mentioned

Teamwork & collaboration · Communication · Problem solving & root cause · Supply chain, planning & logistics

Full posting text

A job at TSMC Arizona offers an opportunity to work at the most advanced semiconductor fab in the United States. TSMC Arizona’s first fab will operate it’s leading-edge semiconductor process technology (N4 process), starting production in the first half of 2025. The second fab will utilize its leading edge N3 and N2 process technology and be operational in 2028. The recently announced third fab will manufacture chips using 2nm or even more advanced process technology, with production starting by the end of the decade. America’s leading technology companies are ready to rely on TSMC Arizona for the next generations of chips that will power the digital future.

As a Material Engineer, you will be an integral part of our team at the Nano-Materials Center, advancing the frontiers of nano-materials in the semiconductor industry. Your primary objective will be to manage and improve the raw materials used in fabrication processes, ensuring the highest quality, reliability, and efficiency. This role offers an exceptional

opportunity to apply your technical expertise and problem-solving skills in a dynamic and innovative environment.

You will collaborate closely with suppliers and cross-functional teams to facilitate seamless communication and integration of materials. By sustaining day-to-day operations, troubleshooting equipment issues, and enhancing process efficiency, you will play a key role in maintaining the industry-leading standards of our organization.

Job Description

Your main responsibilities include

Manage Raw Material Issues: Solve raw material issues using a strong technical background to ensure optimal performance and reliability. Sustain Day-to-Day Operations: Maintain day-to-day raw material operations to ensure they meet quality, reliability, and timely delivery standards. Plan and Execute Analysis Projects: Plan and carry out analysis and defect detection projects to identify and resolve material issues. Conduct Supplier Audits: Visit and audit suppliers to assess their materials and processes, ensuring they meet quality standards and reliability requirements. Facilitate Internal Cooperation: Coordinate with internal departments to support the seamless execution of projects and operations. Collaborate on International Projects: Work with cross-function team and suppliers and prepare for overseas engagements. Cooperate closely with headquarters and other international fabs. Perform On-Site Operations at TSMC Facility: Engage in on-site operational activities at TSMC facilities to observe and analyze processes. Collaborate with facility management and staff to address and resolve any technical or operational issues.

Qualifications

Applicants must be legally eligible to work in the United States and have:

Bachelor, Master or Ph.D. degree in Chemistry, Materials Science, Electrical Engineering, Chemical Engineering, Mechanical Engineering, Physics, or Optics and related fields Understanding of semiconductor fabrication processes; material supply chain Knowledge of chemistry, material science and engineering principles Proficiency in operating and troubleshooting high-tech equipment Ability to conduct thorough audits and assessments Skills in project management, especially related to chemical and material processes Experience in enhancing equipment efficiency through technical improvements At least 1 year experience in semiconductor industry as Process Engineer is a strong add

Interpersonal Skills

Good Communication and Interpersonal Skills: Effective communication and interpersonal skills that enable you to work on cross-functional and geographically dispersed teams. Adaptability: Ability to adapt priorities and responsibilities to support business needs. Hands-on Participation: A hands-on participant with a strong sense of ownership. Team Player with Leadership Skills: An excellent team player with leadership skills, capable of multitasking and thriving in a dynamic, fast-paced environment. Goal-Driven: Demonstrated ability to work in a high-performing team culture with a goal-driven mindset. Quick Assimilation: Quick assimilation to new technology. Problem-Solving: Ability to troubleshoot complex problems and to own the efforts to address and resolve root causes.

Shift Expectations

Standard work hours are Monday through Friday, and occasional weekend shifts.

Candidates must be willing and able to work on-site at our Phoenix Arizona facility.

Work Location : Phoenix, AZ

Travel : May require travel dependent on business needs

TSMC is an equal opportunity employer. We celebrate diversity and are committed to creating an inclusive environment for all employees. All qualified applicants will receive consideration for employment without regard to race, color, religion, age, sex, sexual orientation, gender identity, national origin, disability, veteran status, or any other protected characteristic. We encourage all qualified individuals to apply, and we welcome applications from individuals with diverse backgrounds and experiences.